

Document type: Engineering build specification + product definition (single source of truth for implementation)

Version: 2.0 — "Multi-Agent Diagnostic Reasoning" build

Intended readers: Claude Code (primary implementer), engineering, design, clinical advisors, regulatory lead

Status: Ready for build. MVP scope locked. Demo-mode auth enabled.

0. How to use this document (instructions for the AI builder)

This is the authoritative product definition. **Build the entire system described here as a working, runnable application.** Do not stub out core flows — implement them end to end so the product can be demoed live by anyone who signs up with an email and password.

Rules of precedence:

- **MUST** = hard acceptance criterion. **SHOULD** = strongly preferred. **MAY** = optional.
- The **safety architecture (§9)** and **grounding rules** are non-negotiable. If any other requirement (speed, "smarter" output, slicker UX) conflicts with safety/grounding, **safety/grounding wins.**
- Where this document is silent on an implementation detail, choose the simpler, safer, more auditable option and leave a `// DESIGN-DECISION:` comment.
- The product must **feel like consulting a second specialist** — it asks clarifying questions, reasons out loud (in a structured, auditable way), and never just dumps an answer.

Demo posture (important change from v1): This build is for demonstration. There is **no clinician-credential gate**. Anyone may sign up with email + password and immediately use the full system. Every place a "registered practitioner" guard would normally sit, replace it with a lightweight account + a persistent, dismissible banner: *"Demo build — decision-support only, not for real patient care."* All safety, grounding, audit, and anti-bias mechanisms remain fully active; only the credential wall is removed.

1. Product summary

Aether Clinician is a clinician-facing diagnostic and management **decision-support system (CDSS)** built around a **multi-agent diagnostic reasoning engine**. It behaves like a tireless second specialist sitting beside a primary-care doctor: it reads the patient's fragmented history (paper prescriptions, lab PDFs, scanned reports), builds a structured longitudinal record, **asks the clarifying questions a specialist would ask**, and then

produces grounded differential diagnoses, the most discriminating next investigation, drug-safety checks, and guideline-cited management options.

What makes it feel revolutionary is the **orchestrated agent team** under the hood — specialist reasoning agents, a devil's-advocate agent, a verifier, and a "can't-miss" sentinel — that debate and cross-check each other before anything reaches the clinician. Every clinical claim is traceable to either (a) the patient's own data or (b) a cited clinical guideline. The clinician is always the decision-maker. The system is explicitly engineered to make the clinician **reason more, not less**.

2. Problem and rationale

A primary-care doctor in rural or semi-urban India sees high patient volumes with limited time, no specialist on hand, and patients arriving with fragmented paper histories. The gap between a generalist and a specialist is not only knowledge — it is **time, longitudinal context, and pattern access**. Prior records are scattered across paper prescriptions and lab printouts that are never digitized or reasoned over as a whole.

A well-grounded, multi-agent LLM system can raise the floor of diagnostic reasoning toward specialist level and answer the question a generalist often cannot ask of themselves under time pressure — *"what should I check next, and what am I about to miss?"* The documented risk is **automation bias**: clinicians over-trusting confident-but-wrong AI. The entire product is designed to capture the upside while structurally resisting the downside — chiefly by surfacing disagreement, counter-evidence, and uncertainty rather than hiding them.

3. Goals, non-goals, success metrics

3.1 Product goals. Turn fragmented paper/PDF history into a clean, queryable longitudinal record. Run a multi-agent reasoning process that *interrogates the case* like a specialist. Provide grounded, ranked differentials with explicit evidence, disagreement, and uncertainty. Recommend the single most discriminating next investigation. Check drug interactions, contraindications, allergies, and dosing against the full record. Offer guideline-cited management options. Keep the clinician in control and reasoning actively. Be auditable and regulator-ready by design.

3.2 Non-goals (MVP). No autonomous diagnosis or prescription. No patient-facing mode. No wearable ingestion. No third-party clinical APIs other than the chosen LLM provider. No insurance/claims workflows. No diagnostic assertion presented as certainty.

3.3 Success metrics. Ingestion: a standard lab PDF or prescription image structured and confirmable in <30s. Differential: correct diagnosis in top-5 on a held-out validation set at a clinician-set threshold. **Can't-miss recall: weighted highest.** Citation faithfulness: ≥95% of management statements verifiably supported by the cited guideline section.

Human-in-the-loop (the single most important criterion): a generalist using the tool outperforms the same generalist without it *without inheriting the tool's errors*. Auditability: 100% of clinical outputs logged with evidence and clinician accept/override.

4. Scope

4.1 In scope (MVP) — build all of this

Area	Requirement
Account & demo identity	Email + password signup, session auth, per-account audit identity. No clinician-credential gate. Demo banner persistent.
Patient management	Create/select/search patients; manage multiple patients per account
Record ingestion	PDF, image, and camera/file upload of prescriptions and lab reports
Structured extraction	Multimodal LLM/OCR extraction → normalization → validation → clinician confirmation of low-confidence items
Longitudinal patient graph	Encounters, meds (start/stop/dose), lab time-series, diagnoses, conditions, allergies, all source-linked
Interactive intake (specialist questioning)	Agent-driven adaptive questioning: the system asks the highest-information-gain follow-ups before committing to a differential
Multi-agent differential engine	Orchestrated specialist agents + devil's-advocate + can't-miss sentinel + verifier; ranked, evidence-for/against, confidence band
Recommended investigations	Most discriminating next test, with cost/local-availability framing
Drug safety	Interaction, contraindication, allergy (hard block), renal/hepatic dose checks — all from app-owned curated data
Management options	Guideline-RAG-cited options only; never instructions; prescriber-framed

Verifier & autonomy grading	Conservative-wins verifier; Informational / Suggestive / Flag-for-review tiers
Anti-bias UX	Evidence-before-conclusion, mandatory counter-evidence, agent-disagreement display, active-engagement gates
Audit	Immutable log of every suggestion, evidence, agent trace, and clinician decision
Offline tolerance	Record access + rule-based checks offline; AI reasoning degrades gracefully with a clear "AI paused" signal
Export	Clinician-facing patient summary + structured export

4.2 Out of scope (MVP — do not build)

Patient-facing mode; wearables/sensors; ABDM/ABHA auto-pull; external drug or clinical-reference APIs; autonomous diagnosis or prescribing; insurance workflows; multi-clinic admin dashboards; any feature that asserts a diagnosis/management step as certain or allows one-tap acceptance of high-stakes outputs.

5. Core user flow (the experience we are building)

A user signs up with email + password and lands on a clean patient list. They create or select a patient and upload that patient's historical prescriptions and lab reports (drag-drop, file, or camera). The system extracts and structures them, asking the user to confirm any low-confidence fields inline. The user sees the assembled **longitudinal record**: medication timeline, lab trends, conditions, allergies — every datum clickable back to its source document.

The user enters the presenting complaint. **Here the product becomes a specialist**: instead of immediately answering, the reasoning engine runs a short **adaptive intake** — it asks the two or three highest-yield clarifying questions (duration, character, red-flag screens, relevant negatives) chosen to maximally narrow the differential, exactly as a consultant would. As the user answers, a live **"reasoning theatre"** panel shows the agent team forming and revising hypotheses.

The system then returns a **ranked differential**. Each entry shows patient-specific supporting evidence, opposing evidence, a confidence band, and a can't-miss flag where relevant. Critically, where the agents disagreed, the UI shows the disagreement (e.g., *"Cardiology-agent ranks ACS #1; Devil's-advocate notes atypical features and normal prior*

ECG"). Below the differential sits the **single most discriminating next investigation**, framed by cost and local availability.

The user reviews automatic **drug-safety checks** against the full record (allergy conflicts hard-blocked). If they request management guidance, the system returns **only guideline-cited options**, prescriber-framed; if no guideline is retrieved above threshold, it declines to assert and says so. For any high-stakes or low-confidence output, the user must **actively engage** with the reasoning (acknowledge the counter-evidence) before proceeding — no one-tap accept. Every accept/override is logged. The user can export a clinician summary.

6. The multi-agent diagnostic reasoning engine (the core innovation)

This is the heart of the product. Implement it as an **orchestrated graph of specialized agents**, not a single prompt. Use a stateful agent-orchestration approach (a LangGraph-style state graph is the recommended pattern: agents are nodes, shared case-state flows through edges, conditional edges drive the loop). If you prefer a different orchestrator, that's a **// DESIGN-DECISION**: — but the **roles, the debate, and the conservative-wins verifier are mandatory**.

Shared state object (**CaseState**) flows through every node and carries: the patient graph snapshot, presenting complaint, intake Q&A so far, the current hypothesis set (each with evidence-for, evidence-against, probability band, can't-miss flag), retrieved guideline chunks, agent messages/trace, verifier verdicts, and the autonomy tier.

The agents (nodes):

1. **Triage/Intake Agent.** Decides whether enough information exists to reason. If not, it generates the **highest-information-gain clarifying questions** (target: questions that best split the current hypothesis set, including deliberate red-flag screens and relevant negatives). It loops with the user until information gain per question drops below a threshold or a question cap is hit. This is what makes the product *ask more questions like a specialist*.
2. **Hypothesis Agents (specialist panel).** A small panel of role-primed reasoning agents (e.g., general-internal-medicine, cardiology, infectious-disease, plus a "common things are common" primary-care agent). Each proposes/ranks candidate diagnoses **with explicit patient-data evidence for and against**, drawing only from the patient graph and retrieved guidelines. Run them in parallel where the orchestrator allows.
3. **Can't-Miss Sentinel.** A dedicated agent whose sole job is to scan for dangerous, time-critical, or commonly-missed conditions and force them onto the differential **even at low probability**. Its flags cannot be silently dropped by ranking.
4. **Devil's-Advocate Agent.** Adversarially attacks the leading hypothesis: finds disconfirming evidence, alternative explanations, and base-rate problems. Its output

is **always shown to the clinician** (this is a primary anti-automation-bias mechanism — the user sees the strongest case *against* the top answer).

5. **Investigation Strategist.** Given the contested hypothesis set, computes the **single most discriminating next test**(the one whose result most changes posterior probabilities), framed with approximate cost and local availability.
6. **Guideline-RAG Agent.** For management questions, retrieves from the curated guideline corpus (§8) and drafts options **strictly grounded in retrieved chunks**, each with a citation. If retrieval is below threshold, it returns "insufficient guideline support — declining to assert."
7. **Verifier Agent (gatekeeper).** Independently re-checks every candidate output against (a) the patient data and (b) the retrieved guidelines using a self-consistency / cross-checking pass. **When the reasoning agents and the verifier disagree, the more conservative output wins.** The verifier also assigns the **autonomy tier** based jointly on confidence and clinical stakes. The verifier **cannot be bypassed**.
8. **Synthesis/Orchestrator.** Reconciles the panel, the sentinel, the devil's-advocate, and the verifier into the final structured output, preserving disagreement rather than smoothing it over, and writes the full trace to the audit log.

Tool use available to agents (function-calling tools, app-owned — no external clinical APIs):

- `query_patient_graph(filter)` — structured read of the patient's record.
- `get_lab_trend(marker)` — time-series for a lab marker.
- `check_drug_safety(drug, patient_id)` — deterministic interaction/contraindication/allergy/renal-hepatic check against curated data.
- `retrieve_guidelines(query, k)` — vector retrieval over the versioned guideline corpus, returns chunks with source + section IDs.
- `compute_discriminating_test(hypotheses)` — ranks candidate investigations by expected information gain.
- `flag_cant_miss(symptoms, context)` — returns red-flag/can't-miss conditions to force onto the differential.

Determinism & reproducibility. Every agent run records model version, prompt version, tool calls, and retrieved chunk IDs into the trace, so any output can be reproduced and audited. The deterministic drug-safety tool is the source of truth for safety blocks — agents may *explain* a block but may never *override* it.

7. Data model (guidance for the builder)

Model a **Patient** with one-to-many: **Document** (original file + extraction metadata + per-field confidence), **Encounter**, **MedicationEvent** (drug ref, dose, start/stop/change, source link), **LabResult** (marker, value, unit, reference range, timestamp, source link), **Condition**, **Allergy**, **DerivedMarker** (e.g., eGFR computed from creatinine).

Model **ClinicalSuggestion** capturing: output type (differential / investigation / management / safety-flag), autonomy tier, confidence band, linked evidence (patient-data refs and/or guideline citations), the **full agent trace** (which agents said what, including dissent), verifier verdict, and the clinician decision (accept / override / edit) with timestamp and account id. **Every ClinicalSuggestion is immutable once logged.**

Drug names resolve through a **DrugVocabulary** table mapping Indian brand names → generic → reference identifier, built on a curated open base and localized.

Model **IntakeQuestion** / **IntakeAnswer** so the adaptive-questioning dialogue is part of the auditable record.

8. Guideline corpus & RAG (grounding for management)

Build a **curated, versioned guideline corpus** seeded with India-relevant primary-care content — ICMR **Standard Treatment Workflows (STWs)** as the spine, supplemented by applicable WHO and NICE content where STWs are silent. Each document is chunked with **stable section identifiers** so citations point to a specific section, not a whole document. Store corpus **version** alongside each chunk; **ClinicalSuggestion** records which corpus version it cited.

RAG rules: management text **MUST** be generated only from retrieved chunks; **every management statement MUST carry a specific guideline + section citation**; if retrieval confidence is below threshold, the system **declines to answer** rather than free-generating. Target ≥95% citation faithfulness (each management claim verifiably supported by its cited section).

*(For the demo, seed the corpus with a representative subset of STWs sufficient to make common primary-care presentations work end to end, and clearly mark the corpus as a demo subset with a // **DESIGN-DECISION**: noting how to expand it.)*

9. Safety architecture (non-negotiable)

Every clinical output **MUST** pass through the **verifier** before display; conservative output wins on disagreement; the verifier cannot be bypassed. Each output is tagged with an **autonomy tier**:

- **Informational** (low stakes) — surfaced as facts/links.
- **Suggestive** (moderate) — ranked options with full evidence for and against.
- **Flag-for-review** (low confidence or high stakes) — presented as an *open question* requiring **active clinician engagement** (the clinician must acknowledge the counter-evidence before proceeding).

Low-confidence outputs MUST be rendered as **uncertainty, not fact**. Allergy and hard-contraindication conflicts MUST **hard-block** conflicting wording. No diagnosis or management step is ever presented as certainty. The clinician signs off on every clinical output; every accept/override is logged immutably.

Anti-automation-bias requirements (core, not polish): the UI MUST present **evidence and confidence before the conclusion**; MUST always show the **strongest counter-evidence** and the **devil's-advocate dissent**; MUST require **active engagement** (no one-tap accept) for Flag-for-review outputs; MUST log agreement/override rates for monitoring. The UX MUST NOT use confident phrasing for uncertain outputs and MUST NOT be optimized in any way that encourages blind acceptance.

10. UI/UX specification (build this, don't just imply it)

Design language: clean, calm, clinical; high-contrast, fast, keyboard-friendly; works on a modest laptop. Default light theme with a legible serif/sans pairing; generous whitespace; no gimmicks. Persistent demo banner top of app.

Screens to build:

1. **Auth** — email/password signup + login, minimal friction, "Demo build" note.
2. **Patient List / Home** — searchable list, "New patient," recent activity, big "Start a consult" affordance.
3. **Patient Record (Longitudinal View)** — three regions: (a) **Medication timeline** (horizontal, start/stop/dose-change markers); (b) **Lab trends** (sparkline + expandable charts per marker, abnormal values highlighted, reference ranges shown); (c) **Conditions & allergies** rail. Every datum has a "source" chip linking to the original document with the extracted region highlighted.
4. **Ingestion / Upload** — drag-drop + camera; live extraction progress; **confirmation cards** for low-confidence fields (show the cropped source snippet next to the proposed value; user confirms/edits).
5. **Consult / Reasoning Theatre** — the signature screen. Left: presenting complaint + the **adaptive intake** conversation (the system's clarifying questions appear one at a time, with a subtle "why I'm asking" tooltip). Center: a **live reasoning panel** that shows the agent team's evolving hypothesis list with confidence bars updating as answers come in (this is what makes it *feel* like a specialist thinking). Right: the **dissent feed** (devil's-advocate + can't-miss sentinel notes).
6. **Differential Output** — ranked cards. Each card: diagnosis name, confidence band (as a band, never a single false-precise %), **Evidence For** and **Evidence Against** (both expanded by default — evidence before conclusion), can't-miss badge where relevant, and an **agent-disagreement strip** when the panel split. Below the list: the **"Next best test"** panel with cost + local-availability framing and a one-line rationale ("this most distinguishes A from B").
7. **Drug Safety** — automatic check results; allergy/contraindication conflicts shown as **hard-blocked** red states with explanation; dose concerns flagged with the relevant lab (e.g., low eGFR).

8. **Management Options (Phase 3)** — only guideline-cited cards; each option carries an inline citation chip ("ICMR STW: Hypertension §3.2"); if none retrieved: a clear "Insufficient guideline support — no recommendation" state.
9. **Sign-off & Audit** — for Flag-for-review items, an **active-engagement modal** that requires the clinician to acknowledge the top counter-evidence before accept; accept/override buttons; everything logged.
10. **Export** — clinician summary (current meds, recent abnormalities, key trends, working differential + clinician notes, recommended next steps) as PDF + structured JSON.

Microcopy principle throughout: never imperative ("Give drug X"); always prescriber-framed and evidence-anchored ("Guidelines support considering X; note contraindication Y").

11. Non-functional requirements

Runs on a modest laptop with intermittent internet. Cached record views render <1s; standard extraction <30s. **Core record access and rule-based safety checks MUST work fully offline.** LLM reasoning and RAG require connectivity and MUST **degrade gracefully** with a clear **"AI reasoning paused — offline"** signal; never silently produce degraded reasoning. The only permitted external dependency is the chosen LLM provider.

12. Privacy, security, regulatory posture

All patient health information MUST remain in the India region. Encrypt at rest and in transit; encrypt on-device storage; no plaintext keys. Mandatory consent capture per patient. Immutable audit logs to a compliance-suitable standard. Align with India's **DPDP Act** for sensitive health data.

Regulatory: software supporting diagnosis/treatment for clinical use falls under **CDSCO's Software as a Medical Device (SaMD)** framework (expected Class B/C), requiring registration, a QMS, clinical validation, and post-market surveillance; teleconsultation use must respect the **Telemedicine Practice Guidelines (2020)**. The build MUST produce, from day one, the artifacts a SaMD submission needs: traceability of every clinical output to its evidence, a complete audit trail (including agent traces), versioned guideline corpora, and exportable validation logs. **Regulatory readiness is a build deliverable, not an afterthought.** *(The demo build removes only the credential wall for demonstration; all compliance scaffolding stays in place.)*

13. Clinical validation plan (build the hooks now)

Before any real-patient deployment, validate on held-out real cases for: top-k differential accuracy, **can't-miss recall (weighted highest)**, interaction-flag precision/recall, citation

faithfulness, and the **human-in-the-loop outcome**(generalist-with-tool vs without, confirming clinicians do not inherit the tool's errors). Instrument the app to capture these metrics from real usage logs, including **agent agreement/override rates** and verifier-overrule frequency. "Worked in a demo" is explicitly **not** an acceptance bar for real-patient use.

14. Phasing

Phase 1: account/auth (email+password), ingestion, longitudinal patient graph, deterministic interaction/contraindication/allergy checks, audit log. Shippable and useful alone.

Phase 2: the multi-agent reasoning engine (intake agent, hypothesis panel, can't-miss sentinel, devil's-advocate, investigation strategist), verifier + autonomy grading, reasoning-theatre UI, anti-bias UX, export.

Phase 3: guideline RAG + cited management options.

Phase 4: validation instrumentation maturity, CDSCO pathway artifacts, monitored pilot.

Do not build Phase 2 reasoning before the Phase 1 foundation is solid; do not deploy to real patients before Phase 4 validation.

15. Recommended technical stack (guidance — substitute with // **DESIGN-DECISION**: if you deviate)

Backend application server with a typed schema; relational store for the patient graph + immutable audit (append-only suggestion log); a vector store for the guideline corpus. Agent orchestration via a **stateful graph orchestrator**(LangGraph-style) wrapping the chosen LLM provider with function-calling tools as defined in §6. Multimodal extraction via the provider's vision capability with an OCR fallback for low-quality scans, then a normalization/validation pass. Frontend as a responsive single-page app implementing the screens in §10, with the reasoning-theatre panel driven by streamed agent state. All clinical reasoning behind the verifier gate; all safety blocks behind the deterministic tool.

16. Acceptance criteria (MVP)

The MVP is complete when **any user who signs up with email + password** can: create/select a patient; upload multiple historical prescriptions and lab reports and confirm low-confidence extractions; view a longitudinal record with source-linked lab trends and a medication timeline; enter a presenting complaint and be **asked specialist-style clarifying questions** by the intake agent; watch the **multi-agent reasoning** produce a ranked differential with evidence for/against, confidence bands, can't-miss flags, and **visible agent disagreement / devil's-advocate dissent**; receive the **single most discriminating next**

investigation with cost/availability framing; receive interaction/contraindication/allergy checks against the full record (allergy = hard block); (Phase 3) receive **guideline-cited management options** with **no uncited management text**; be required to **actively engage** with high-stakes outputs rather than one-tap accept; have every suggestion, agent trace, and accept/override **logged immutably**; export a clinician summary; and do all of this with the record and rule-based checks **fully functional offline** and no external dependency beyond the LLM.

Non-negotiables. Clinician is always the decision-maker. No uncited management text. No patient-specific claim without a source link. Allergy conflicts hard-block. Verifier cannot be bypassed. No diagnosis or management presented as certainty. Devil's-advocate dissent and can't-miss flags always surfaced. Download-all and audit-export never fail. No external APIs beyond the LLM. Demo build shows the safety banner at all times.

That's the complete single document — hand it straight to Claude Code and it has everything it needs to build the full product, UI included.

Two honest flags before you run with it.

First, on the "anyone signs up and starts using it" decision: that's exactly right for a *demo*, and I've built it in cleanly (credential wall removed, every safety/grounding/audit mechanism kept). I'd just keep the demo banner non-negotiable, because the one thing that will sink a Government of India conversation is any impression that an un-credentialed person could treat the output as a clinical decision rather than support. The banner is cheap insurance.

Second — and I'll repeat my earlier recommendation because it still holds — **this is a build spec, not your government pitch**. It's now even more build-heavy (multi-agent internals, data models, acceptance tests), which is great for Claude Code and wrong for a minister's office. For the government meeting you want a separate, tighter artifact emphasizing public-health impact at scale, alignment with **eSanjeevani / Ayushman Bharat Digital Mission**, cost-of-care reduction, a phased pilot proposal, and your team/governance story — with the multi-agent + SaMD-readiness angle as your differentiator against the usual "AI diagnoses!" pitches they're wary of.